

Lecture 20: Self-gravitation and rotation

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Outline and motivation

In previous lectures we have discussed how ~~that~~ travel time and waveform observations can be used to determine variations in elastic wave speeds within the Earth. Nothing has, however, been said about **density**, with this parameter being key to understanding both the Earth's interior composition and dynamics. To learn about density, we need to study the ~~long-period free oscillations~~ long-period free oscillations of the Earth. This is because **self-gravitation**, and hence density, plays a vital role in their physics. Within this lecture we will return to basic elastodynamics, and show how gravitational forces can be included into the equations of motion. We also explain how the **Earth's rotation** enters into the problem. Having obtained the equations of motion, we perform a **scaling analysis** to determine the relative importance of the different terms, and conclude that at short enough periods the effects of self-gravitation and rotation can be safely neglected, and hence the earlier parts of this course remain valid.

A review of Newtonian gravitation

Consider the motion φ of an elastic body relative to a reference body M , and write M_t for the instantaneous volume of space occupied at time t . At this instant in time, the gravitational potential of the body satisfies Poisson~~equation~~

$$\nabla^2 \phi = 4\pi G \varrho.$$

's equation

$$\nabla^2 \phi = 4\pi G \varrho, \tag{1}$$

in $\mathbb{R}^3$, where G is the universal gravitational constant, and ϱ is the density. The gravitational acceleration associated with this potential is given by

$$g_i = - \frac{\partial \phi}{\partial x_i}$$
$$\nabla^2 \phi = 4\pi G \varrho. \tag{2}$$

~~The above Poisson~~Poisson's equation is subject to the conditions that ϕ tends to zero at infinity, while across any jumps in density we have the continuity conditions

$$\begin{aligned} \underline{[\phi]_-^+} &= 0, \quad \underline{[g_i \hat{n}_i]_-^+} = 0 \\ \underline{[\phi]_-^+} &= 0, \quad \underline{[g_i \hat{n}_i]_-^+} = 0, \end{aligned} \quad (3)$$

where $[\cdot]_-^+$ denotes the jump in a quantity in the direction of ~~the~~ outward normal. The closed form solution to this problem is given by

$$\begin{aligned} \phi(\mathbf{y}, t) &= -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)}{[(y_i - y'_i)(y_i - y'_i)]^{\frac{1}{2}}} d^3 \mathbf{y}', \\ \underline{\phi(\mathbf{y}, t)} &= -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)}{[(y_i - y'_i)(y_i - y'_i)]^{\frac{1}{2}}} d^3 \mathbf{y}', \end{aligned} \quad (4)$$

and differentiating, we ~~similar find~~

$$\underline{g_i(\mathbf{y}, t)} = -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)(y_i - y'_i)}{[(y_j - y'_j)(y_j - y'_j)]^{\frac{3}{2}}} d^3 \mathbf{y}'.$$

~~similarly find~~

$$\underline{g_i(\mathbf{y}, t)} = -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)(y_i - y'_i)}{[(y_j - y'_j)(y_j - y'_j)]^{\frac{3}{2}}} d^3 \mathbf{y}'. \quad (5)$$

For the Earth, the equilibrium gravitational acceleration and the pressure that it produces are shown in Fig. 1. These were calculated from the radial density profile of the model PREM; where that density comes from is a question we return to in Lecture 24, since it is the free oscillations discussed there that provide the main constraint on it. Note that g is close to 10 m s^{-2} throughout the mantle, reaching its largest value at the core-mantle boundary, and decreases roughly linearly through the core.

It will be convenient to define the **referential gravitational potential** by

$$\begin{aligned} \underline{\zeta(\mathbf{x}, t)} &= \phi[\varphi(\mathbf{x}, t), t], \\ \underline{\zeta(\mathbf{x}, t)} &= \phi[\varphi(\mathbf{x}, t), t], \end{aligned} \quad (6)$$

which is a time-dependent field on the reference body, M . We recall from Lecture ~~12~~ 13 that the volume elements in the reference body and physical space are related by $\underline{d^3 \mathbf{y}} = J(\mathbf{x}, t) d\mathbf{x}$ ~~$d^3 \mathbf{y} = J(\mathbf{x}, t) d^3 \mathbf{x}$~~ with $J = \det(\mathbf{F})$ the Jacobian of the motion. We can then change variables within eq. ~~(4)~~ (4) and use the definition of ζ in eq. ~~(6)~~ (6) to obtain

$$\underline{\zeta(\mathbf{x}, t)} = -G \int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)][\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]\}^{\frac{1}{2}}} d\mathbf{x}',$$

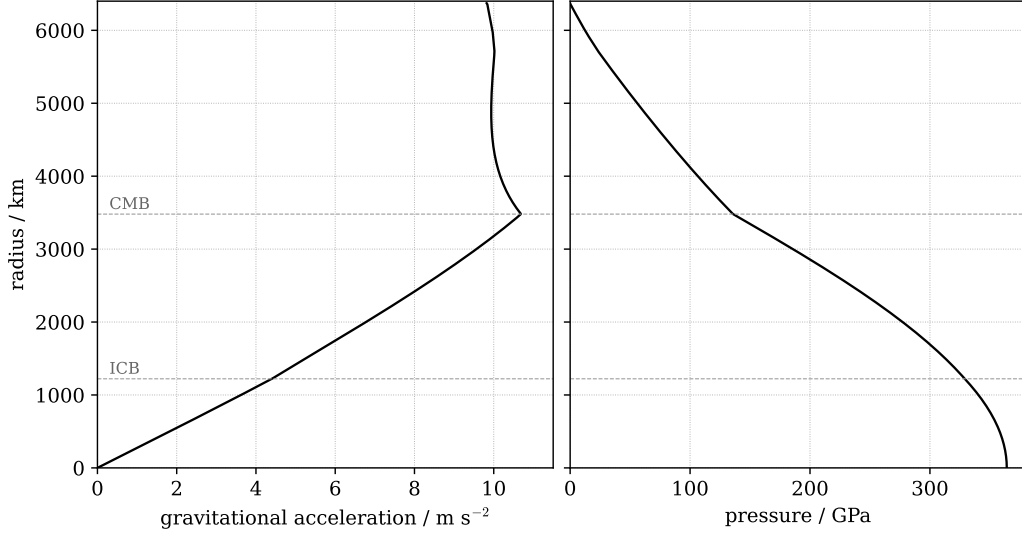


Fig. 1: Gravitational acceleration (left) and pressure (right) as functions of radius in the earth model PREM, calculated from its density profile. The dashed lines mark the core–mantle boundary (CMB) and inner core boundary (ICB).

$$\zeta(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)][\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]\}^{\frac{1}{2}}} d^3\mathbf{x}', \quad (7)$$

where from Lecture 12–13 we know that conservation of mass requires

$$\rho(\mathbf{x}) = J(\mathbf{x}, t) \varrho[\varphi(\mathbf{x}, t), t], \quad (8)$$

with ρ the **referential density**. In a similar manner, we define the **referential gravitational acceleration** by

$$\gamma_i(\mathbf{x}, t) = g_i[\varphi(\mathbf{x}, t), t], \quad (9)$$

which from eq. (5) can be written

$$\gamma_i(\mathbf{x}, t) = -G \int_M \frac{\rho(\mathbf{x}') [\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]}{\{[\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)][\varphi_j(\mathbf{x}, t) - \varphi_j(\mathbf{x}', t)]\}^{\frac{3}{2}}} d^3\mathbf{x}', \quad (10)$$

Gravitational binding energy

At each instant of time we can associate a **gravitational binding energy** with the elastic body. This is the energy that would be required to assemble the body from matter dispersed at infinity, assuming that its constituent parts are acted on only by their

mutual gravitational attraction. Because gravity is a conservative force, the binding energy is independent of the details of the assembly.

Suppose that the mass of the body is already in place, and that its spatial density and gravitational potential are denoted by ρ and ϕ . To determine an expression for the binding energy we will consider how this quantity changes due to an infinitesimal addition of mass to the body. By definition of the gravitational potential, the work required to bring in an additional mass element $\delta\rho d^3\mathbf{y}$ from infinity to a point $\mathbf{y}$ within the body is $\delta\rho \phi d^3\mathbf{y}$. The ~~total change in the~~ binding energy associated with a perturbation $\delta\rho$ to ~~the~~ body's density ~~by~~ is, therefore, given by

$$\begin{aligned} \delta\mathcal{V}_g &= \int_{M_t} \delta\rho \phi d^3\mathbf{y}. \\ \delta\mathcal{V}_g &= \int_{M_t} \delta\rho \phi d^3\mathbf{y}, \end{aligned} \quad (11)$$

to first-order accuracy. Using the perturbed Poisson's equation $\nabla^2\delta\phi = 4\pi G\delta\rho$ for the resulting change in gravitational potential, we can write this increment to the binding energy as

$$\begin{aligned} \delta\mathcal{V}_g &= \frac{1}{4\pi G} \int_{M_t} \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y} = \frac{1}{4\pi G} \int_{\mathbb{R}^3} \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y}, \\ \delta\mathcal{V}_g &= \frac{1}{4\pi G} \int_{M_t} \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y} = \frac{1}{4\pi G} \int_{\mathbb{R}^3} \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi d^3\mathbf{y}, \end{aligned} \quad (12)$$

where the final equality follows from the fact that $\nabla^2\delta\phi = 0$ outside of the instantaneous body M_t . Using the identity

$$\begin{aligned} \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi &= \frac{\partial}{\partial x_i} \left(\phi \frac{\partial\delta\phi}{\partial x_i} \right) - \frac{\partial\phi}{\partial x_i} \frac{\partial\delta\phi}{\partial x_i} \\ \left(\frac{\partial^2\delta\phi}{\partial x_i \partial x_i} \right) \phi &= \frac{\partial}{\partial x_i} \left(\phi \frac{\partial\delta\phi}{\partial x_i} \right) - \frac{\partial\phi}{\partial x_i} \frac{\partial\delta\phi}{\partial x_i}, \end{aligned} \quad (13)$$

and applying the divergence theorem we obtain

$$\begin{aligned} \delta\mathcal{V}_g &= -\frac{1}{4\pi G} \int_{\mathbb{R}^3} \frac{\partial\phi}{\partial x_i} \frac{\partial\delta\phi}{\partial x_i} d^3\mathbf{y}. \\ \delta\mathcal{V}_g &= -\frac{1}{4\pi G} \int_{\mathbb{R}^3} \frac{\partial\phi}{\partial x_i} \frac{\partial\delta\phi}{\partial x_i} d^3\mathbf{y}, \end{aligned} \quad (14)$$

where we have used the continuity of the potentials and their normal derivatives across ∂M_t , while the surface integral at infinity vanishes because $\phi \sim r^{-1}$ and $\|\nabla\delta\phi\| \sim r^{-2}$ as $r \rightarrow \infty$. It follows that the increment to the gravitational binding energy can be written

$$\delta\mathcal{V}_g = \delta \left(-\frac{1}{8\pi G} \int_{\mathbb{R}^3} \frac{\partial\phi}{\partial x_i} \frac{\partial\phi}{\partial x_i} d^3\mathbf{y} \right)$$

$$\delta \mathcal{V}_g = \delta \left(-\frac{1}{8\pi G} \int_{\mathbb{R}^3} \frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} d^3 \mathbf{y} \right), \quad (15)$$

and so we arrive at the result

$$\mathcal{V}_g = -\frac{1}{8\pi G} \int_{\mathbb{R}^3} \frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} d^3 \mathbf{y}. \quad (16)$$

Using ~~the Poisson~~ Poisson's equation for ϕ , we can write the integrand of the above expression in the form

$$\frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \phi}{\partial x_i} \right) - \phi \left(\frac{\partial^2 \phi}{\partial x_i \partial x_i} \right) = \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \phi}{\partial x_i} \right) - 4\pi G \varrho \phi, \quad (17)$$

and applying the divergence theorem we obtain

$$\mathcal{V}_g(t) = \frac{1}{2} \int_{M_t} \varrho(\mathbf{y}, t) \phi(\mathbf{y}, t) d^3 \mathbf{y}, \quad (18)$$

where, as above, we have used the continuity of ϕ and its normal derivative across ∂M_t , and the surface integral at infinity again vanishes. To include this binding energy into Hamilton's principle, it will be necessary to transform eq. ~~(18)~~ (18) to the following integral over the reference body

$$\mathcal{V}_g(t) = \frac{1}{2} \int_M \rho(\mathbf{x}) \zeta(\mathbf{x}, t) d^3 \mathbf{x}, \quad (19)$$

where we have used the definition of the referential gravitational potential in eq. ~~(6)~~ (6) and the statement of conservation of mass during the deformation in eq. ~~(8)~~ (8).

The equations of motion relative to inertial space

Having obtained an appropriate expression for the gravitational binding energy, the equations of motion can be readily obtained using Hamilton's principle. The Lagrangian for the problem is

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2} \rho(\mathbf{x}) \|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2} \rho(\mathbf{x}) \zeta(\mathbf{x}, t),$$

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2}\rho(\mathbf{x})\zeta(\mathbf{x}, t), \quad (20)$$

where we note that the elastic strain energy is allowed to be time-dependent to accommodate a seismic source. An important point about this Lagrangian is that it is **non-local** due to the gravitational binding energy. This means that the value of the Lagrangian at a point and time cannot be expressed only in terms of the values of the various fields and their derivatives at this point. The physical reason for this non-locality is, of course, the **action-at-a-distance** property of Newtonian gravitation.

The action for the body takes the usual form

$$\mathcal{S}(\varphi) = \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \varphi(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] d\mathbf{x} dt, \quad (21)$$

and Hamilton's principle requires that this quantity be stationary about the true motion for all perturbations with fixed end points. In obtaining the Euler-Lagrange equations, however, we have to account for the non-local nature of the Lagrangian. As ever, the mathematical statement of Hamilton's principle is ~~that the~~ vanishing of the functional derivative of the action

$$\langle DS(\varphi) | \delta\varphi \rangle = 0, \quad (22)$$

for all variations, $\delta\varphi$, subject to the usual fixed-endpoint conditions. Here the ~~the left hand~~ side is, by definition, just the linear term in the expansion of $\mathcal{S}(\varphi + \delta\varphi)$ about φ . Looking at the form of the action, it is clear that this linear term takes the form

$$\langle DS(\varphi) | \delta\varphi \rangle = \int_0^T \int_M \left(\frac{\delta\mathcal{L}}{\delta\varphi_i} \delta\varphi_i + \frac{\delta\mathcal{L}}{\delta v_i} \delta v_i + \frac{\delta\mathcal{L}}{\delta F_{ij}} \delta F_{ij} \right) d\mathbf{x} dt. \quad (23)$$

where we have introduced the convenient notation $\frac{\delta\mathcal{L}}{\delta\varphi_i}$, $\frac{\delta\mathcal{L}}{\delta v_i}$ and $\frac{\delta\mathcal{L}}{\delta F_{ij}}$ for the factors multiplying the various perturbations. In the case of a local Lagrangian these factors reduce to normal partial derivatives, and hence it is only the term $\frac{\delta\mathcal{L}}{\delta\varphi_i}$ that requires additional thought. Applying the usual integration by parts steps, we arrive at the ~~Euler-Lagrange equation~~

$$\frac{\delta\mathcal{L}}{\delta\varphi_i} - \frac{\partial}{\partial t} \left(\frac{\delta\mathcal{L}}{\delta v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\delta\mathcal{L}}{\delta F_{ij}} \right) = 0,$$

Euler-Lagrange equation

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} - \frac{\partial}{\partial t} \left(\frac{\delta \mathcal{L}}{\delta v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\delta \mathcal{L}}{\delta F_{ij}} \right) = 0, \quad (24)$$

along with the natural boundary condition

$$\frac{\delta \mathcal{L}}{\delta F_{ij}} \hat{n}_j = 0$$

$$\frac{\delta \mathcal{L}}{\delta F_{ij}} \hat{n}_j = 0 \quad (25)$$

on ∂M .

As in the non-gravitating case we have

$$\frac{\delta \mathcal{L}}{\delta v_i} = \rho v_i, \quad \frac{\delta \mathcal{L}}{\delta F_{ij}} = -\frac{\partial W}{\partial F_{ij}}$$

$$\frac{\delta \mathcal{L}}{\delta v_i} = \rho v_i, \quad \frac{\delta \mathcal{L}}{\delta F_{ij}} = -\frac{\partial W}{\partial F_{ij}}, \quad (26)$$

while in the ~~final~~second problem set you will show that

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho \gamma_i$$

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho \gamma_i, \quad (27)$$

with γ_i the referential gravitational acceleration as defined above. The equations of motion, therefore, take the expected form

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} - \rho \gamma_i = 0,$$

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} - \rho \gamma_i = 0, \quad (28)$$

with $T_{ij} = \frac{\partial W}{\partial F_{ij}}$ the first ~~Piola Kirchhoff~~Piola-Kirchhoff stress tensor. The form of these equations is not, of course, unexpected. To the balance of linear momentum obtained previously, we have simply added the gravitational force acting on the body due to its own mass.

Use of a co-rotating reference frame

We have obtained the equations of motion for a self-gravitating body with respect to an **inertial reference** frame defined by the “fixed stars”, and these results could be applied directly to the Earth. As, however, the Earth undergoes a nearly steady rotation about its polar axis, it is useful to reformulate the equations of motion with respect to a **non-inertial frame that rotates with the Earth**.

To do this we express the motion as

$$\begin{aligned} \varphi_i(\mathbf{x}, t) &= \Phi_i(t) + R_{ij}(t) \bar{\varphi}_j(\mathbf{x}, t), \\ \varphi_i(\mathbf{x}, t) &= \Phi_i(t) + R_{ij}(t) \bar{\varphi}_j(\mathbf{x}, t), \end{aligned} \quad (29)$$

where $\mathbf{R}(t)$ is a rotation matrix. Here ~~$\Phi(t)$ denotes~~ $\Phi(t)$ denotes the translation of the reference frame relative to inertial space, and $\mathbf{R}(t)$ changes of orientation. The term $\bar{\varphi}(\mathbf{x}, t)$ describes the motion relative to the non-inertial frame, and will be called the **relative motion**.

Differentiating eq. (29) with respect to time we see that the velocity relative to inertial space can be written

$$\begin{aligned} v_i &= \frac{d\Phi_i}{dt} + R_{ij} \left(\frac{\partial \bar{\varphi}_j}{\partial t} + R_{kl} \frac{dR_{kl}}{dt} \bar{\varphi}_l \right) \\ v_i &= \frac{d\Phi_i}{dt} + R_{ij} \left(\frac{\partial \bar{\varphi}_j}{\partial t} + R_{kl} \frac{dR_{kl}}{dt} \bar{\varphi}_l \right), \end{aligned} \quad (30)$$

where we have used the identity $R_{kj}R_{kl} = \delta_{jl}$ for rotation matrices. In fact, if we differentiate this latter identity we find

$$\begin{aligned} \frac{dR_{kj}}{dt} R_{kl} + R_{kj} \frac{dR_{kl}}{dt} &= 0. \\ \frac{dR_{kj}}{dt} R_{kl} + R_{kj} \frac{dR_{kl}}{dt} &= 0, \end{aligned} \quad (31)$$

and hence conclude that $R_{kj} \frac{dR_{kl}}{dt}$ is an **anti-symmetric matrix**. It follows that we can use the alternating tensor to write

$$\begin{aligned} R_{kj} \frac{dR_{kl}}{dt} &= \epsilon_{jkl} \Omega_k \\ R_{kj} \frac{dR_{kl}}{dt} &= \epsilon_{jml} \Omega_m, \end{aligned} \quad (32)$$

for a uniquely determined vector Ω . Using this result, the velocity becomes

$$v_i = \frac{d\Phi_i}{dt} + R_{ij} (\bar{v}_j + \epsilon_{jkl} \Omega_k \bar{\varphi}_l).$$

$$\underline{v_i} = \frac{d\Phi_i}{dt} + R_{ij}(\bar{v}_j + \epsilon_{jkl}\Omega_k\bar{\varphi}_l). \quad (33)$$

Here $\bar{\mathbf{v}}$ is the velocity as seen from the non-inertial frame, while $\underline{\Omega}$ is the **angular velocity of the frame**. Differentiating again and applying the same ideas, the acceleration relative to inertial space can be written

$$\begin{aligned} \frac{\partial v_i}{\partial t} &= \frac{d^2\Phi_i}{dt^2} + R_{ij} \left(\frac{\partial \bar{v}_j}{\partial t} + 2\epsilon_{jkl}\Omega_k\bar{v}_l + \epsilon_{jkl}\frac{d\Omega_k}{dt}\bar{\varphi}_l + \epsilon_{jkl}\epsilon_{lmn}\Omega_k\Omega_m\bar{\varphi}_n \right) \\ \underline{\frac{\partial v_i}{\partial t}} &= \underline{\frac{d^2\Phi_i}{dt^2} + R_{ij} \left(\frac{\partial \bar{v}_j}{\partial t} + 2\epsilon_{jkl}\Omega_k\bar{v}_l + \epsilon_{jkl}\frac{d\Omega_k}{dt}\bar{\varphi}_l + \epsilon_{jkl}\epsilon_{lmn}\Omega_k\Omega_m\bar{\varphi}_n \right)}. \end{aligned} \quad (34)$$

On the **right hand side** we can identify various terms. The first is the acceleration of the frame relative to inertial space. Within the brackets we then see, relative to the non-inertial frame, accelerations associated with the motion along with the expected **Coriolis**, **Euler**, and **centrifugal** contributions.

We have not yet specified *how* the non-inertial frame is to be chosen. Within seismology the most common approach is to take

$$\begin{aligned} \underline{\frac{d\Phi}{dt}} &= \underline{0}, \quad \underline{\frac{d\Omega}{dt}} = \underline{0}, \\ \underline{\frac{d\Phi}{dt}} &= \underline{0}, \quad \underline{\frac{d\Omega}{dt}} = \underline{0}, \end{aligned} \quad (35)$$

which corresponds to a frame whose **origin is static** relative to the **“fixed stars”** but **rotates steadily**, with the angular velocity chosen so that the frame **approximately co-rotates with the Earth**. The advantage of this choice is its simplicity. But it does mean that orbital, rotational, and relative motion of the Earth are mixed up within the term $\bar{\varphi}$. A preferable approach is to use the so-called **Tisserand frame** which is defined such that the relative motion, $\bar{\varphi}$ carries no net linear **nor angular momentum**¹. **or angular momentum**¹. But we will stick to the simpler and more common choice in this course.

To complete the derivation of the equations of motion in the non-inertial frame, we note from eq. (29) that

$$\underline{F_{ij}} = \underline{R_{ik}\bar{F}_{kj}}$$

(29) that

$$\underline{F_{ij}} = \underline{R_{ik}\bar{F}_{kj}}, \quad (36)$$

with $\bar{\mathbf{F}}$ the deformation gradient associated with relative motion. Due to material frame indifference, we then have $W(\mathbf{x}, \mathbf{F}) = W(\mathbf{x}, \bar{\mathbf{F}})$ as should be physically expected. Using the chain rule, we find

$$\underline{\bar{T}_{ij}} = \frac{\partial W}{\partial \bar{F}_{ij}}(\bar{\mathbf{F}}) = \frac{\partial}{\partial \bar{F}_{ij}} W(\mathbf{R}\bar{\mathbf{F}}) = \frac{\partial W}{\partial F_{kl}}(\mathbf{R}\bar{\mathbf{F}}) \frac{\partial}{\partial \bar{F}_{ij}} (R_{km}\bar{F}_{ml}) = R_{ki}T_{kj}$$

¹ See “On the elastodynamics of rotating planets” by Maitra & Al-Attar (2024) for details.

¹ See Maitra & Al-Attar (2024) for details.

$$\bar{T}_{ij} = \frac{\partial W}{\partial \bar{F}_{ij}}(\bar{\mathbf{F}}) = \frac{\partial}{\partial \bar{F}_{ij}} W(\mathbf{R}\bar{\mathbf{F}}) = \frac{\partial W}{\partial F_{kl}}(\mathbf{R}\bar{\mathbf{F}}) \frac{\partial}{\partial \bar{F}_{ij}} (R_{km} \bar{F}_{ml}) = R_{ki} T_{kj}, \quad (37)$$

with the ~~left hand~~ left-hand side the stress associated with $\bar{\varphi}$. Passing the rotation matrix to the other side, we have then obtained the useful identity

$$\begin{aligned} T_{ij} &= R_{ik} \bar{T}_{kj} \\ T_{ij} &= R_{ik} \bar{T}_{kj}, \end{aligned} \quad (38)$$

between the first ~~Piola-Kirchhoff~~ Piola-Kirchhoff stress relative to the inertial and non-inertial reference frames. Finally, we put eq.(29 (29)) into eq.(10 (10)) to find the relationship

$$\begin{aligned} \gamma_i &= R_{ij} \bar{\gamma}_i \\ \gamma_i &= R_{ij} \bar{\gamma}_j, \end{aligned} \quad (39)$$

between the referential gravitational acceleration in the two frames, with ~~$\bar{\gamma}$~~ $\bar{\gamma}$ defined exactly as in eq.(10 (10)) but with $\bar{\varphi}$ replacing φ . Using these various results we have shown that the equations of motion are equivalent to

$$\begin{aligned} \rho \left(\frac{\partial \bar{v}_i}{\partial t} + 2\epsilon_{ijk} \Omega_j \bar{v}_k + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l \bar{\varphi}_m \right) - \frac{\partial \bar{T}_{ij}}{\partial x_j} - \rho \bar{\gamma}_i &= 0. \\ \rho \left(\frac{\partial \bar{v}_i}{\partial t} + 2\epsilon_{ijk} \Omega_j \bar{v}_k + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l \bar{\varphi}_m \right) - \frac{\partial \bar{T}_{ij}}{\partial x_j} - \rho \bar{\gamma}_i &= 0. \end{aligned} \quad (40)$$

These equations are identical in form to those in an inertial frame except for the inclusion of the Coriolis and centrifugal accelerations – the Euler acceleration has been removed because we have restricted attention to a steadily rotating frame. From this point on we will consistently work in the approximately co-rotating frame, and hence will simplify notations by dropping overlines on terms linked to the relative motion.

Linearised equations of motion

As ever, our main concern is with the linearised equations of motion about an equilibrium state. Strictly, we should use the term **relative equilibrium**, as the body is now undergoing a ~~steadily~~ steady rotation relative to inertial space. We consider a disturbance to the body caused by a stress glut, and hence take the strain energy to be

$$\begin{aligned} W(\mathbf{x}, t, \mathbf{F}) &= U(\mathbf{x}, \mathbf{C}) - s \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij} \\ W(\mathbf{x}, t, \mathbf{F}) &= U(\mathbf{x}, \mathbf{C}) - s \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij}, \end{aligned} \quad (41)$$

with s a perturbation parameter, and $\bar{\mathbf{S}}$ a given stress glut. As in Lecture 15, the elastic tensor that appears below is the second derivative of the time-independent part

$U(\mathbf{x}, \mathbf{C})$ of this strain energy function. In a now standard manner we can obtain linearised equations of motion for the displacement vector defined as the first-order term in the expansion

$$\begin{aligned} \varphi_i(\mathbf{x}, t) &= x_i + s u_i(\mathbf{x}, t) + O(s^2), \\ \varphi_i(\mathbf{x}, t) &= x_i + s u_i(\mathbf{x}, t) + O(s^2). \end{aligned} \quad (42)$$

At ~~zeroth-order~~ zeroth order, the equations of motion take the form

$$\begin{aligned} \rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 &= 0, \\ \rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 &= 0, \end{aligned} \quad (43)$$

where T_{ij}^0 and γ_i^0 denote the equilibrium values of the stress tensor and gravitational acceleration. We will have more to say about these equations in the following lecture, but for the moment note that there must be a balance between the centrifugal ~~and~~ and gravitational forces on the body and ~~a~~ the resulting equilibrium stress; the simplifying assumption of a stress-free equilibrium is no longer an option.

From the first-order term in the expansion we arrive at the linearised equations of motion for the displacement field

$$\begin{aligned} \rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho \gamma_i^1 &= -\frac{\partial \bar{S}_{ij}}{\partial x_j}, \\ \rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho \gamma_i^1 &= -\frac{\partial \bar{S}_{ij}}{\partial x_j}, \end{aligned} \quad (44)$$

where γ_i^1 denotes the first-order perturbation to the referential gravitational ~~potential~~ acceleration. Using a binomial expansion, this term can be written explicitly as

$$\begin{aligned} \gamma_i^1 &= -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d\mathbf{x}', \\ \gamma_i^1 &= -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d^3\mathbf{x}', \end{aligned} \quad (45)$$

where it is understood that primed terms are functions of the integration variable. The boundary conditions for the problem take the expected form

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j.$$

$$\underbrace{A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j}_{\text{action-at-a-distance}} = \bar{S}_{ij} \hat{n}_j. \quad (46)$$

The key point is that the inclusion of gravitation and rotation into the problem only requires some additional terms in the equations of motion considered so far. It is notable, however, that these are no longer partial differential equations, but **integro-partial-differential-integro-differential equations**, with both derivatives and integrals of the unknown functions occurring within the equations of motion. Again, this feature derives from the **action-at-a-distance** property of Newtonian gravitation.

The neglect of self-gravitation and rotation

We have obtained the linearised equations of motion within a steadily rotating and self-gravitating body. It is natural to ask if and why it has been reasonable to neglect these physical phenomena up till this point. To do so, we will perform a scaling analysis of the equations.

We assume that the displacement vector field is characterised by a typical amplitude $\|\mathbf{u}\| \sim U$, a length-scale $\|\mathbf{u}\| \sim U$, a length scale L and time-scale T . The following **order-of-magnitude** scaling estimates are then immediate

$$\begin{aligned} \underbrace{\rho \frac{\partial^2 u_i}{\partial t^2}} &\sim \underbrace{\frac{\bar{\rho} U}{T^2}}, \\ \underbrace{\rho \epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t}} &\sim \underbrace{\frac{\bar{\rho} \Omega U}{T}}, \\ \underbrace{\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m} &\sim \underbrace{2 \bar{\rho} \Omega^2 U}, \\ \underbrace{\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right)} &\sim \underbrace{\frac{\bar{\mu} U}{L^2}}, \end{aligned}$$

$$\underbrace{\rho \frac{\partial^2 u_i}{\partial t^2}} \sim \underbrace{\frac{\bar{\rho} U}{T^2}}, \quad (47)$$

$$\underbrace{\rho \epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t}} \sim \underbrace{\frac{\bar{\rho} \Omega U}{T}}, \quad (48)$$

$$\underbrace{\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l u_m} \sim \underbrace{\bar{\rho} \Omega^2 U}, \quad (49)$$

$$\underbrace{\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right)} \sim \underbrace{\frac{\bar{\mu} U}{L^2}}, \quad (50)$$

where $\bar{\rho}$ and $\bar{\mu}$ denote typical values of density and elastic **moduli** within the body, while Ω is the magnitude of the rotation vector. The scaling estimate for the gravitational term within the equations of motion requires more thought. The referential

gravitational potential at equilibrium is given by

$$\zeta_0 = -G \int_M \frac{\rho'}{[(x_k - x'_k)(x_k - x'_k)]^{1/2}} d\mathbf{x}'.$$

$$\zeta_0 = -G \int_M \frac{\rho'}{[(x_k - x'_k)(x_k - x'_k)]^{1/2}} d^3\mathbf{x}'. \quad (51)$$

Taking the gradient of this expression twice, we obtain

$$\frac{\partial \zeta_0}{\partial x_i \partial x_j} = G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) d\mathbf{x}',$$

$$\frac{\partial^2 \zeta_0}{\partial x_i \partial x_j} = G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) d^3\mathbf{x}', \quad (52)$$

and using ~~the Poisson~~Poisson's equation $\nabla^2 \zeta_0 = 4\pi G \rho$ along with eq. ~~(45)~~(45) we arrive at the desired scaling estimate

$$\rho \gamma_i^1 \sim 4\pi G \bar{\rho}^2 U$$

$$\rho \gamma_i^1 \sim 4\pi G \bar{\rho}^2 U. \quad (53)$$

Using these results, we can quantify the relative importance of the terms within the force balance. First we have

$$\frac{\text{gravitational forces}}{\text{elastic forces}} = \frac{4\pi G \bar{\rho}^2 U}{\frac{\bar{\mu} U}{L^2}} = \frac{4\pi G \bar{\rho} L^2}{\bar{v}^2},$$

$$\frac{\text{gravitational forces}}{\text{elastic forces}} = \frac{4\pi G \bar{\rho}^2 U}{\frac{\bar{\mu} U}{L^2}} = \frac{4\pi G \bar{\rho} L^2}{\bar{v}^2}, \quad (54)$$

where $\bar{v} = \sqrt{\frac{\bar{\mu}}{\bar{\rho}}}$ $\bar{v} = \sqrt{\bar{\mu}/\bar{\rho}}$ represents a typical elastic wave speed in the model. This ratio scales like L^2 where L is the ~~characteristics length-scale~~characteristic length scale of the deformation. We conclude that for deformations occurring over sufficiently small ~~length-scales~~length scales, gravitational forces will be negligible relative to elastic ones. We should then ask whether gravitational forces will ever be of importance within the Earth? To answer this, we note that the largest-scale deformation within the Earth can have L equal to its average radius ~~6371 km~~6371 km. Recalling that $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, and taking as typical values $\bar{\rho} = 5000 \text{ kg m}^{-3}$ and $\bar{v} = 8000 \text{ ms}^{-1}$ we find

$$\frac{\text{gravitational forces}}{\text{elastic forces}} \sim 3.$$

$\bar{\rho} = 5000 \text{ kg m}^{-3}$ and $\bar{v} = 8000 \text{ m s}^{-1}$, we find

$$\frac{\text{gravitational forces}}{\text{elastic forces}} \sim 3. \quad (55)$$

For such planetary scale deformations, it follows that gravitational forces are of the same order of magnitude as elastic forces, and so cannot be neglected within the dynamics.

Next, we consider the ratio of the Coriolis and inertial terms, and find

$$\frac{\text{Coriolis term}}{\text{inertial term}} = \frac{\frac{\bar{\rho}\Omega U}{T}}{\frac{\bar{\rho}U}{T^2}} = \Omega T$$

$$\frac{\text{Coriolis term}}{\text{inertial term}} = \frac{\frac{\bar{\rho}\Omega U}{T}}{\frac{\bar{\rho}U}{T^2}} = \Omega T. \quad (56)$$

It follows that the Coriolis terms will be negligible for motions whose **time-scale** time scale is short relative to the rotation period of the Earth. A similar conclusion is readily shown to hold for the centrifugal term. It is known that seismic deformation is limited to periods shorter than about one hour, and so rotational effects play a relatively small (but not negligible) role in the problem. There are, however, processes at longer periods, e.g. those associated with solid-Earth tides, for which rotation becomes very important.

The conclusions of this scaling analysis can be checked directly by computing synthetic seismograms in a spherically symmetric earth model with and without self-gravitation, as shown in Fig. 2, where attenuation has been neglected so that the long-period waves persist over the six hours shown.² Over the full frequency band of the calculation, which extends to 50 mHz, the two seismograms are indistinguishable, and they remain so when band-pass filtered to periods shorter than a few minutes, in keeping with the estimate that gravitational forces are negligible for deformations on length scales much smaller than the radius of the Earth. As the pass band is moved to longer periods, however, differences appear and steadily grow, and in the lowest band, at periods between eight and seventeen minutes, they are substantial: the oscillations are slightly but systematically slower without gravitation, so that the two waveforms drift apart as the long-period surface waves circle the Earth. These are precisely the deformations whose length scale is comparable to that of the planet, and we shall see in Lecture 23 that the periods of the corresponding free oscillations of the Earth are altered by self-gravitation.

What you need to know and be able to do

- (i) Understand the relevant parts of Newtonian gravity, including the manner in which the gravitational potential and acceleration can be transformed into quantities defined on the reference body.

² The calculations use the program `yspec`, which solves the equations of motion of this lecture in a spherically symmetric earth model by direct numerical integration in the frequency domain (Al-Attar & Woodhouse 2008).

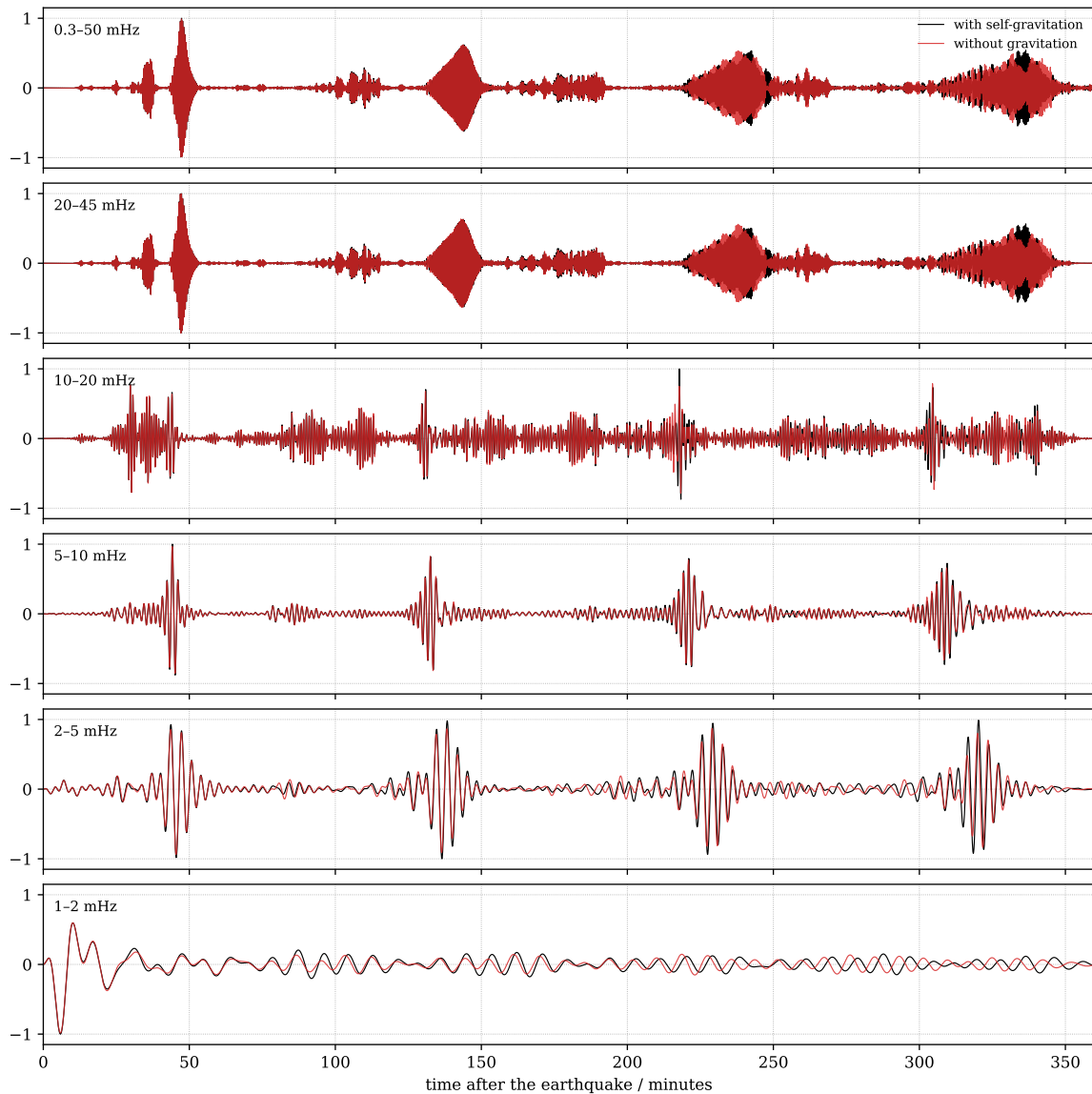


Fig. 2: Synthetic vertical-component acceleration seismograms in the earth model PREM for an earthquake of moment magnitude 8 at 20 km depth, recorded at an epicentral angle of 90° , computed with (black) and without (red) self-gravitation. The top trace shows the full band of the calculation, from 0.3 to 50 mHz, and those below it the same seismogram band-pass filtered in successively lower frequency bands. Each trace is normalised by its maximum, and a taper has been applied over the first ten minutes to suppress a transient at the origin time. The calculations neglect attenuation, so that the surface waves continue to circle the Earth undiminished over the six hours shown. The two calculations differ only in the lowest bands, and there the differences grow with time as the long-period surface waves circle the Earth.

- (ii) Know physically what the gravitational binding energy of a planet represents, and how it can be determined.
- (iii) Know how the gravitational binding energy is incorporated into the Lagrangian, and how the equations of motion can be obtained. The full derivation is too long to be examined, but you may be asked to reproduce parts of it given appropriate information and guidance.
- (iv) Perform scaling arguments to determine the relative size of terms ~~with-in~~ within an equation, and to determine whether some of them might be negligible. This is a technique required throughout this course, and you should ~~insure~~ ensure you are happy with it.

Lecture 21: Equilibrium figures

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Outline and motivation

In the previous lecture we obtained the equations of motion for a self-gravitating and rotating earth model, and showed that these effects are relevant at sufficiently long periods. A particular conclusion was that the equilibrium state of such an earth model **cannot be stress-free** as has been assumed in earlier parts of the course. In this lecture we investigate the constraints imposed by the equilibrium equations in further detail, and show that they are key to understanding the **observed shapes of planets**.

The equilibrium equations

In the previous lecture we obtained the following equilibrium equations

$$\begin{aligned} \rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 &= 0, \\ \rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 &= 0, \end{aligned} \tag{1}$$

for a self-gravitating and steadily rotating planet¹. Here ρ is the density, Ω_i the angular velocity, T_{ij}^0 the stress, and γ_i^0 the gravitational acceleration. To simplify ~~notations~~notation, let us write σ_{ij} for the equilibrium stress, and g_i for the gravitational acceleration. The equilibrium equations then become

$$\begin{aligned} \rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial \sigma_{ij}}{\partial x_j} - \rho g_i &= 0, \\ \rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial \sigma_{ij}}{\partial x_j} - \rho g_i &= 0, \end{aligned} \tag{2}$$

¹ ~~Though our focus in this course is on the Earth, the material of this lecture applies so naturally to other planets it seems sensible to generalise the discussion somewhat.~~

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which are subject to the boundary conditions $\sigma_{ij}\hat{n}_j = 0$ on ∂M . Though these equations were derived from elasticity, the force balance expressed is valid in more general materials. This is important because over long ~~time-scales~~ time scales the Earth or Earth-like planets are not elastic.

If we knew the shape, density distribution, and angular velocity of the planet, then all terms in the equilibrium equations ~~are~~ would be fixed except for the stress tensor. Recalling from Lecture ~~13-14~~ 13-14 that equilibrium stress tensors are necessarily symmetric, we see that we have only three equations for six unknown functions. The problem is, therefore, ~~under-determined~~ underdetermined, and we cannot expect to uniquely determine ~~stress-tensor~~² the stress tensor.²

Hydrostatic equilibrium

You have seen in the first part of this course that over thousands of years the interior of the Earth flows like a fluid, this being due to solid-state creep. Considering for simplicity a Newtonian fluid, we know that the stress tensor³ takes the form

$$\sigma_{ij} = -p \delta_{ij} + \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3} \frac{\partial v_k}{\partial x_k} \delta_{ij} \right)$$

$$\sigma_{ij} = -p \delta_{ij} + \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3} \frac{\partial v_k}{\partial x_k} \delta_{ij} \right), \quad (3)$$

where p is the pressure, η the viscosity, and v_i the velocity. Note that the given expression is valid in a **compressible fluid**, but it reduces to that for an incompressible one by requiring the velocity to be divergence-free. If the fluid is not flowing, then the stress tensor takes the simple hydrostatic form

$$\sigma_{ij} = -p \delta_{ij}$$

$$\sigma_{ij} = -p \delta_{ij}, \quad (4)$$

and the same is true for non-Newtonian fluids. Such an expression must also be approximately valid if the fluid is flowing sufficiently slowly. To get a sense of this approximation for the Earth, we note that the size of the non-hydrostatic stress associated with mantle convection scales like $\eta V/L$ with $\eta \sim 10^{21} \text{ Pa s}$ $\eta \sim 10^{21} \text{ Pa s}$ a typical

² ~~This counting argument is suggestive but not entirely correct. Rather, the relevant point is that there exist non-zero symmetric tensor fields having vanishing divergence. This can be compared to divergence-free vector fields with which you will be very familiar (i.e. divergence of a curl is zero). In fact, a method for generating divergence-free tensor fields exists using an extension of the curl operator that generates divergence-free symmetric tensor fields.~~

² This counting argument is suggestive but not entirely correct. Rather, the relevant point is that there exist non-zero symmetric tensor fields having vanishing divergence. This can be compared to divergence-free vector fields with which you will be very familiar (i.e. the divergence of a curl is zero). In fact, a method for generating divergence-free tensor fields exists using an extension of the curl operator that generates divergence-free symmetric tensor fields.

³ This is the Cauchy stress that occurs within spatial formulations of continuum mechanics. But note that at equilibrium the referential and spatial pictures coincide due to our choice of reference body.

viscosity, $V \sim 10^{-8} \text{ m s}^{-1}$ $V \sim 10^{-8} \text{ m s}^{-1}$ a typical speed of a few centimetres per year, and $L \sim 10^6 \text{ m}$ $L \sim 10^6 \text{ m}$ the length scale. From this we find that deviatoric stresses are around $\eta V/L \sim 10^7 \text{ Pa}$ $\eta V/L \sim 10^7 \text{ Pa}$. Pressures in the mantle are, however, known to be of order 10^{10} Pa 10^{10} Pa ,⁴ and hence the stress is quite close to being hydrostatic.

This motivates us to look for **hydrostatic** solutions of the equilibrium equations. Putting eq.(4) into eq.(2) we obtain

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m + \frac{\partial p}{\partial x_i} - \rho g_i = 0.$$

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m + \frac{\partial p}{\partial x_i} - \rho g_i = 0, \quad (5)$$

along with the boundary condition $p = 0$ on ∂M . In assuming the stress is hydrostatic we have moved from an **under-determined** ~~underdetermined~~ problem to an **over-determined** ~~overdetermined~~ one, with three equations for a single unknown, p . The form of this hydrostatic equilibrium equation can be usefully simplified by noting first that

$$g_i = - \frac{\partial \phi}{\partial x_i}$$

$$g_i = - \frac{\partial \phi}{\partial x_i}, \quad (6)$$

with ϕ the equilibrium gravitational potential. Similarly, the centrifugal term can also be written as the gradient of a potential

$$\epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m = \frac{\partial \psi}{\partial x_i}$$

$$\epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m = \frac{\partial \psi}{\partial x_i}, \quad (7)$$

where we have defined

$$\psi = \frac{1}{2} (\Omega_i \Omega_j - \Omega_k \Omega_k \delta_{ij}) x_i x_j$$

$$\psi = \frac{1}{2} (\Omega_i \Omega_j - \Omega_k \Omega_k \delta_{ij}) x_i x_j. \quad (8)$$

Moving to vector notation, the hydrostatic equilibrium equation takes the simple form

$$\nabla p + \rho \nabla \gamma = \mathbf{0},$$

$$\nabla p + \rho \nabla \gamma = \mathbf{0}, \quad (9)$$

⁴ Values for the pressure can be calculated once the density profile is known.

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where we have introduced the **gravity potential**, $\gamma = \phi + \psi$. Note that “gravity” is used here for the **combined effect of gravitational and centrifugal forces**.

An immediate consequence of eq.(9) is that ∇p and $\nabla \gamma$ are parallel, and hence the level surfaces of these fields are coincident. In particular, pressure vanishes on the surface of the planet, and so this must also be an equipotential of gravity. If we take the curl of eq.(9), recalling $\nabla \times \nabla f = 0$ for any scalar field, we find

$$\begin{aligned} \nabla \rho \times \nabla \gamma &= 0, \\ \nabla \rho \times \nabla \gamma &= 0, \end{aligned} \tag{10}$$

and so ~~note that~~ ρ and γ also have coincident level surfaces. In conclusion we have seen that for ~~an~~ a planet to be in hydrostatic equilibrium the level surfaces of p , ρ , and γ must all be the same, as illustrated in Fig. 1.

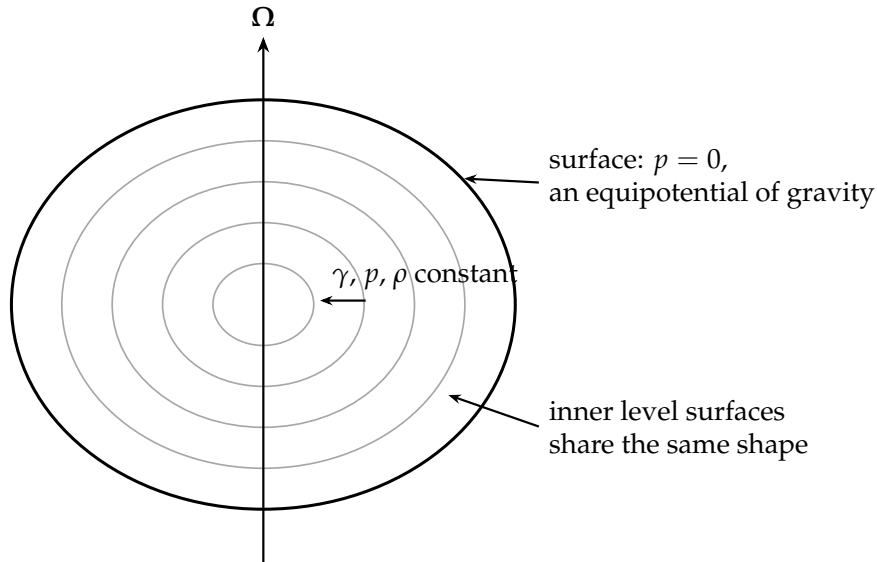


Fig. 1: In a planet in hydrostatic equilibrium the level surfaces of the pressure, the density and the gravity potential all coincide. The outer surface, on which the pressure vanishes, is itself an equipotential of gravity, and every interior level surface has the same character.

Suppose we arbitrarily specified the shape, density variations, and angular velocity of ~~an~~ a planet. We could then, in principle, determine its gravitational potential by solving Poisson’s equation, and hence find the gravity potential. There would be no reason, however, that this field would have level surfaces that agree with those for the chosen density. This suggests that in requiring a planet to be in hydrostatic equilibrium we are placing rather strong restrictions on, and interrelations between, its shape, density distribution, and angular velocity.

Non-rotating hydrostatic planets

As a starting point, suppose that the planet does not rotate, and hence $\psi = 0$. Assume also that the planet is spherical in shape, and that density varies only with radius. We

then know that ϕ is also spherically symmetric, and hence eq.(9) reduces to

$$\begin{aligned} \nabla p + \rho \frac{d\phi}{dr} \hat{r} &= 0, \\ \nabla p + \rho \frac{d\phi}{dr} \hat{r} &= 0, \end{aligned} \quad (11)$$

with $\hat{r}$ a radial unit vector. To solve this equation we can clearly take p to be a function of r alone, and hence need only integrate

$$\begin{aligned} \frac{dp}{dr} + \rho \frac{d\phi}{dr} &= 0;, \\ \frac{dp}{dr} + \rho \frac{d\phi}{dr} &= 0, \end{aligned} \quad (12)$$

using the boundary condition $p = 0$ on the surface $r = b$. This argument shows that for a non-rotating planets-planet it is possible to find at least one class of hydrostatic equilibrium states, namely those that are spherically symmetric. From a physical perspective, it seems reasonable that these are the only hydrostatic equilibria possible. This is indeed true, but a mathematical proof of this result is not easy^{5,5}.

Slowly rotating hydrostatic planets

Given that a non-rotating and hydrostatic planet must be spherically symmetric, we would expect that a slowly rotating planet should have an equilibrium figure that is close to spherical. In this section, we show that this is indeed the case. First, however, we should clarify what is meant by **slowly rotating**.

The essential idea is that the ratio of the centrifugal to gravitational potentials should be small. Looking at these quantities on the planet's surface we have approximately

$$\begin{aligned} \psi &\sim \frac{1}{2} \Omega^2 b^2 \\ \psi &\sim \frac{1}{2} \Omega^2 b^2, \end{aligned} \quad (13)$$

where Ω is the angular frequency and b the mean radius, while also

$$\phi \sim \frac{4}{3} \pi G \bar{\rho} b^2$$

⁵ In fact, I am aware of a proof only in the case of a homogeneous planet, with the argument summarised in Chandrasekhar's book "Ellipsoidal figures of equilibrium".

⁵ The case of a homogeneous planet is summarised in the book by Chandrasekhar (1969). The general result, that a non-rotating fluid body in hydrostatic equilibrium under its own gravitation must be spherically symmetric, is due to Lichtenstein (1918, 1933). It can also be seen as an instance of a more general symmetry property of solutions of quasi-linear elliptic equations, which is what the hydrostatic equations amount to once the pressure has been eliminated in favour of the potential.

$$\phi \sim \frac{4}{3} \pi G \bar{\rho} b^2, \quad (14)$$

with $\bar{\rho}$ the mean density. The ratio of these terms is given by

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim \frac{3\Omega^2}{8\pi G \bar{\rho}}$$

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim \frac{3\Omega^2}{8\pi G \bar{\rho}}. \quad (15)$$

For the Earth, we have $\bar{\rho} \sim 5000 \text{ kg m}^{-3}$ and $\Omega \sim 7 \times 10^{-5} \text{ s}^{-1}$ $\bar{\rho} \sim 5000 \text{ kg m}^{-3}$ and $\Omega \sim 7 \times 10^{-5} \text{ s}^{-1}$, and hence we find

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim 2 \times 10^{-3},$$

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim 2 \times 10^{-3}, \quad (16)$$

which is indeed a fairly small quantity.

We model the rotating planet through a first-order perturbation of a spherical reference model. The density in the reference model will be written ρ_0 , and similarly for other quantities. Within the rotating planet we then have

$$\begin{aligned} \underline{\rho} &= \rho_0 + s \rho_1 + \dots, \\ \underline{p} &= p_0 + s p_1 + \dots, \\ \underline{\phi} &= \phi_0 + s \phi_1 + \dots, \\ \underline{\psi} &= s \psi_1, \end{aligned}$$

$$\underline{\rho} = \rho_0 + s \rho_1 + \dots, \quad (17)$$

$$\underline{p} = p_0 + s p_1 + \dots, \quad (18)$$

$$\underline{\phi} = \phi_0 + s \phi_1 + \dots, \quad (19)$$

$$\underline{\psi} = s \psi_1, \quad (20)$$

with s a perturbation parameter introduced for book-keeping purposes. We must also consider the aspherical shape of the planet. For simplicity we neglect internal boundaries^{6,6} and so need only parameterise the outer surface in the form

$$\underline{r(\theta, \varphi)} = b + s h_1(\theta, \varphi) + \dots,$$

⁶ But these can be easily included into the theory.

⁶ But these can be easily included into the theory.

$$\underline{r(\theta, \varphi) = b + s h_1(\theta, \varphi) + \dots}, \quad (21)$$

with (θ, φ) spherical polar co-ordinates.

Putting the above expansions into the equilibrium equations we find at ~~first-order~~

$$\underline{\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0}},$$

first order

$$\underline{\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0}}, \quad (22)$$

where $\gamma_1 = \phi_1 + \psi_1$, and we have written $g_0 = \frac{d\phi_0}{dr}$ for convenience. The boundary condition on pressure is applied on the perturbed surface, and hence

$$\underline{p[b + s h_1(\theta, \varphi) + \dots, \theta, \varphi] = 0}.$$

$$\underline{p[b + s h_1(\theta, \varphi) + \dots, \theta, \varphi] = 0}. \quad (23)$$

Expanding this out to ~~first-order~~ first order in s , we arrive at

$$\underline{p_1 + \frac{dp_0}{dr} h_1 = 0}$$

$$\underline{p_1 + \frac{dp_0}{dr} h_1 = 0} \quad (24)$$

on the reference surface $r = b$. The zeroth-order equilibrium equation tells us that

$$\underline{\frac{dp_0}{dr} + \rho_0 g_0 = 0},$$

$$\underline{\frac{dp_0}{dr} + \rho_0 g_0 = 0}, \quad (25)$$

and hence the boundary ~~conditions simplify to~~

$$\underline{p_1 - \rho_0 g_0 h_1 = 0}.$$

condition simplifies to

$$\underline{p_1 - \rho_0 g_0 h_1 = 0}. \quad (26)$$

We will show that ~~eq.(22)~~eqs. (22) and (26) allow the density, pressure, and boundary perturbations to be expressed in terms of that for the gravity potential. First, we take the cross product of eq.(22) with the unit vector $\hat{\mathbf{r}}$ to obtain

$$\underline{\hat{\mathbf{r}} \times \nabla p_1 + \hat{\mathbf{r}} \times \rho_0 \nabla \gamma_1 = \mathbf{0}}.$$

$$\underline{\hat{\mathbf{r}} \times \nabla p_1 + \hat{\mathbf{r}} \times \rho_0 \nabla \gamma_1 = \mathbf{0}}. \quad (27)$$

Noting that

$$\begin{aligned} \hat{\mathbf{r}} \times \rho_0 \nabla \gamma_1 &= \hat{\mathbf{r}} \times \left[\nabla(\rho_0 \gamma_1) - \gamma_1 \frac{d\rho_0}{dr} \hat{\mathbf{r}} \right] = \hat{\mathbf{r}} \times \nabla(\rho_0 \gamma_1), \\ \hat{\mathbf{r}} \times \rho_0 \nabla \gamma_1 &= \hat{\mathbf{r}} \times \left[\nabla(\rho_0 \gamma_1) - \gamma_1 \frac{d\rho_0}{dr} \hat{\mathbf{r}} \right] = \hat{\mathbf{r}} \times \nabla(\rho_0 \gamma_1), \end{aligned} \quad (28)$$

eq.(27) can be written

$$\begin{aligned} \hat{\mathbf{r}} \times \nabla(p_1 + \rho_0 \gamma_1) &= \mathbf{0}, \\ \hat{\mathbf{r}} \times \nabla(p_1 + \rho_0 \gamma_1) &= \mathbf{0}. \end{aligned} \quad (29)$$

This relation tells us that the gradient of $p_1 + \rho_0 \gamma_1$ is parallel to $\hat{\mathbf{r}}$, and hence this function depends on radius alone. We are free to assume that the perturbations all average to zero over spherical surfaces^{7,7} and hence we arrive at

$$\begin{aligned} p_1 &= -\rho_0 \gamma_1 \\ p_1 &= -\rho_0 \gamma_1, \end{aligned} \quad (30)$$

which expresses the pressure perturbation in terms of that to the gravity potential. From the perturbed boundary condition we then immediately obtain

$$\begin{aligned} h_1 &= -\frac{\gamma_1}{g_0} \\ h_1 &= -\frac{\gamma_1}{g_0}. \end{aligned} \quad (31)$$

For the second stage of the argument, we take the curl of eq.(22) to obtain

$$\begin{aligned} \frac{d\rho_0}{dr} \hat{\mathbf{r}} \times \nabla \gamma_1 + g_0 \nabla \rho_1 \times \hat{\mathbf{r}} &= \mathbf{0}, \\ \frac{d\rho_0}{dr} \hat{\mathbf{r}} \times \nabla \gamma_1 + g_0 \nabla \rho_1 \times \hat{\mathbf{r}} &= \mathbf{0}. \end{aligned} \quad (32)$$

Using the same trick as before, this is equivalent to

$$\begin{aligned} \hat{\mathbf{r}} \times \nabla \left(\frac{d\rho_0}{dr} \gamma_1 - g_0 \rho_1 \right) &= \mathbf{0}, \\ \hat{\mathbf{r}} \times \nabla \left(\frac{d\rho_0}{dr} \gamma_1 - g_0 \rho_1 \right) &= \mathbf{0}, \end{aligned} \quad (33)$$

⁷ If not, these terms could be combined into the spherical reference model.

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from which we see that the density perturbation is given by

$$\rho_1 = \frac{1}{g_0} \frac{d\rho_0}{dr} \gamma_1$$

$$\rho_1 = \frac{1}{g_0} \frac{d\rho_0}{dr} \gamma_1. \quad (34)$$

To summarise, we started from the hydrostatic equilibrium equations in a slowly rotating planet, and have obtained the following identities

$$p_1 = -\rho_0 \gamma_1, \quad h_1 = -\frac{\gamma_1}{g_0}, \quad \rho_1 = \frac{1}{g_0} \frac{d\rho_0}{dr} \gamma_1.$$

$$p_1 = -\rho_0 \gamma_1, \quad h_1 = -\frac{\gamma_1}{g_0}, \quad \rho_1 = \frac{1}{g_0} \frac{d\rho_0}{dr} \gamma_1. \quad (35)$$

To complete the discussion, we recall that ϕ_1 is also related to ρ_1 and h_1 through solution of Poisson's equation. Indeed, we have

$$\nabla^2 \phi_1 = \begin{cases} 4\pi G \rho_1 & r < b \\ 0 & r > b \end{cases}$$

$$\nabla^2 \phi_1 = \begin{cases} 4\pi G \rho_1 & r < b \\ 0 & r > b \end{cases} \quad (36)$$

while the appropriate continuity conditions at $r = b$ are

$$[\phi_1]_{-}^{+} = 0, \quad [\hat{\mathbf{r}} \cdot \nabla \phi_1 + 4\pi G \rho_0 h_1]_{-}^{+} = 0$$

$$[\phi_1]_{-}^{+} = 0, \quad [\hat{\mathbf{r}} \cdot \nabla \phi_1 + 4\pi G \rho_0 h_1]_{-}^{+} = 0, \quad (37)$$

whose derivation is left as an exercise in the second problem set. Using eq. (35), we see that Poisson's equation is transformed into

$$\nabla^2 \phi_1 = \begin{cases} 4\pi G \frac{1}{g_0} \frac{d\rho_0}{dr} (\phi_1 + \psi_1) & r < b \\ 0 & r > b \end{cases}$$

$$\nabla^2 \phi_1 = \begin{cases} 4\pi G \frac{1}{g_0} \frac{d\rho_0}{dr} (\phi_1 + \psi_1) & r < b \\ 0 & r > b \end{cases} \quad (38)$$

along with the continuity conditions

$$[\phi_1]_{-}^{+} = 0, \quad \left[\hat{\mathbf{r}} \cdot \nabla \phi_1 - 4\pi G \rho_0 \frac{(\phi_1 + \psi_1)}{g_0} \right]_{-}^{+} = 0.$$

$$[\phi_1]_{-}^{+} = 0, \quad \left[\hat{\mathbf{r}} \cdot \nabla \phi_1 - 4\pi G \rho_0 \frac{(\phi_1 + \psi_1)}{g_0} \right]_{-}^{+} = 0. \quad (39)$$

Eq.(38) and (39) constitute a boundary value problem that can be solved for ϕ_1 in terms of ψ_1 . Having done this we can then use eq.(35) to determine the other parameters for the perturbed planet. These equations, albeit in a less general form, were first obtained by Clairaut (1713-1765), this work building on Newton's original discussion of equilibrium figures in the "*Principia*" (Newton 1687). To implement these equations in practice we must know the variation of spherically averaged density with radius, and in later lectures we will see how this was determined for the Earth.

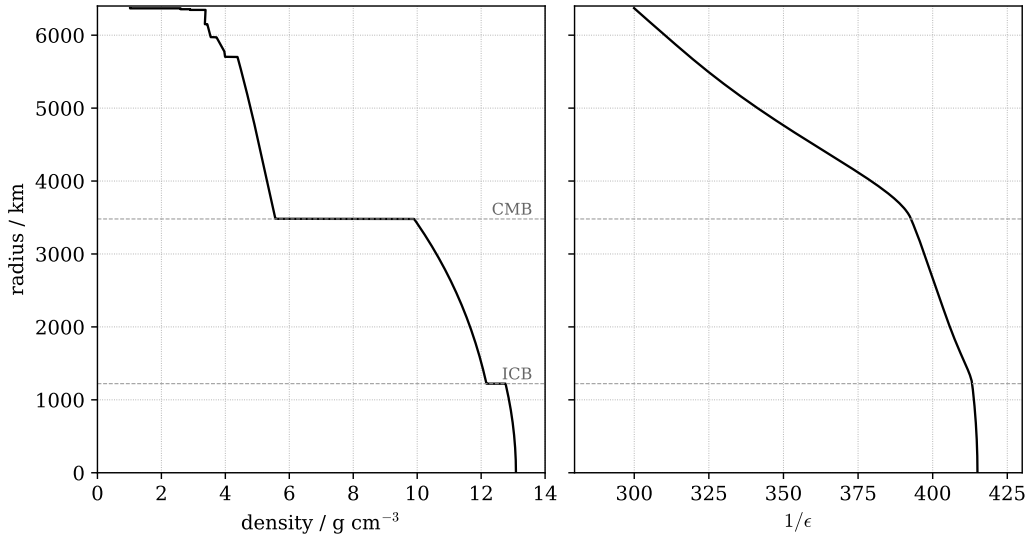


Fig. 2: The upper panel shows variations in Left: the Earth's spherically averaged density structure as according to the model PREM, which was determined from free oscillation data along with the mass and moment of inertia constraints. The lower figure then plots Right: the reciprocal of the ellipticity as determined through obtained by numerical solution of Clairaut's equation for this density profile. At the surface $1/\epsilon \approx 300$, while at the centre it is about 415.

It is worth noting these arguments apply to more general potentials, ψ_1 , and not only that linked to the centrifugal force. The theory can, therefore, also be applied to model things like equilibrium tides within gaseous planets. Another point to make is that the equations are linear in ϕ_1 and ψ_1 , and hence symmetries of ψ_1 are directly mirrored in ϕ_1 and the other perturbed quantities. In terms of spherical harmonics, the aspherical part of the centrifugal potential perturbation can be shown to be proportional to Y_{20} . It is for this reason that equilibrium figures of slowly rotating planets are found to be oblate spheroids, and characterised in terms of a single radial function, ϵ , known as the ellipticity and defined such that

$$\phi_1 + \psi_1 = \sqrt{\frac{4\pi}{5}} \frac{2}{3} r g_0 \epsilon Y_{20}. \quad (40)$$

At the mean surface $r = b$ it can be shown that the ellipticity is such that

$$\epsilon(b) = \frac{\text{equatorial radius} - \text{polar radius}}{\text{mean radius}}$$

$$\epsilon(b) = \frac{\text{equatorial radius} - \text{polar radius}}{\text{mean radius}}, \quad (41)$$

and for the Earth this value is around 1/300. Fig. 2 shows the ellipticity obtained by solving Clairaut's equation, given below, for the Earth's spherically averaged density. The hydrostatic value at the surface is 1/299.9, slightly smaller than the observed flattening of 1/298.3, and this small discrepancy is a measure of the non-hydrostatic stresses supported by the mantle. Using properties of spherical harmonics, it can be shown that eq. (38) reduces to

$$\frac{d^2\epsilon}{dr^2} + 8\pi G\rho_0 g_0^{-1} \left(\frac{d\epsilon}{dr} + \frac{\epsilon}{r} \right) - \frac{6\epsilon}{r^2} = 0.$$

$$\frac{d^2\epsilon}{dr^2} + 8\pi G\rho_0 g_0^{-1} \left(\frac{d\epsilon}{dr} + \frac{\epsilon}{r} \right) - \frac{6\epsilon}{r^2} = 0, \quad (42)$$

which is known as Clairaut's equation and is subject to the boundary conditions

$$\left. \frac{d\epsilon}{dr} \right|_{r=0} = 0, \quad \left. \frac{d\epsilon}{dr} \right|_{r=b} = \frac{1}{b} \left[\frac{5\Omega^2 b^3}{2GM} - 2\epsilon(b) \right]$$

$$\left. \frac{d\epsilon}{dr} \right|_{r=0} = 0, \quad \left. \frac{d\epsilon}{dr} \right|_{r=b} = \frac{1}{b} \left[\frac{5\Omega^2 b^3}{2GM} - 2\epsilon(b) \right], \quad (43)$$

with M the planet's mass. This problem can be solved numerically using a shooting method, whereby the initial conditions at $r = 0$ are progressively adjusted so that the required condition at $r = b$ is met.

Rapidly rotating bodies

The theory above is restricted to slow rotation, but it is worth describing, without any derivations, what is known about the equilibrium figures of a homogeneous fluid body rotating at an arbitrary rate; this material is descriptive, and you will not be asked to reproduce it. The problem has a long history, beginning with Newton and Maclaurin in the eighteenth century, and the results are summarised in the book by Chandrasekhar (1969). For a homogeneous body the natural measure of the rotation rate is the dimensionless ratio $\Omega^2 / \pi G\rho$, which is the quantity estimated in eq. (15) up to a numerical factor, and it is helpful to describe the figures in terms of the angular momentum of the body rather than its rotation rate, since the angular momentum is what is conserved as a body contracts or spins down.

Maclaurin spheroids. Maclaurin showed in 1742 that an oblate spheroid of any eccentricity is an exact equilibrium figure of a homogeneous fluid for a suitable rotation

rate, with $\Omega^2/\pi G\rho$ a function of the eccentricity alone. This function is shown in Fig. 3. It rises from zero for a sphere, in agreement with the slow-rotation theory, but it reaches a maximum of $\Omega^2/\pi G\rho \approx 0.449$ for an eccentricity of about 0.93 and then falls again, so that beyond a certain rotation rate no spheroidal figure exists at all, while below it there are two spheroids for each rotation rate. Along the sequence the angular momentum increases without bound, the highly flattened spheroids rotating slowly but having enormous moments of inertia.

Jacobi ellipsoids. It came as a surprise when Jacobi showed in 1834 that a rotating homogeneous fluid need not be axially symmetric: there is a second family of exact equilibrium figures, the triaxial ellipsoids that now bear his name, which rotate about their shortest axis. As shown in Fig. 3, the Jacobi sequence **bifurcates** from the Maclaurin sequence at an eccentricity of 0.8127, where $\Omega^2/\pi G\rho \approx 0.374$ and the polar axis is 0.58 of the equatorial one. Beyond this point there are therefore two distinct equilibrium figures with the same angular momentum, and the Jacobi ellipsoid has the lower energy. The Maclaurin spheroids become unstable beyond the bifurcation, in the sense that a small triaxial perturbation grows, so long as there is some dissipation to allow the energy to fall, and a slowly contracting fluid body would be expected to pass from the spheroidal to the ellipsoidal sequence as its angular momentum grows. Some members of the two sequences are drawn in Fig. 3.

Pear-shaped figures and fission. Poincaré showed in 1885 that the Jacobi sequence has in turn a point of bifurcation, at an axis ratio $a_2/a_1 \approx 0.43$, from which a sequence of **pear-shaped figures** branches off. It was widely supposed at the time that this sequence might continue to figures with a narrow waist, and finally to two separate bodies, and on this basis George Darwin developed the **fission hypothesis**, in which the Moon was born from a rapidly rotating early Earth and close binary stars from the splitting of a single star. The pear-shaped figures were shown by Liapounov and later by Cartan to be unstable, however, and the fission hypothesis has long been abandoned; the Moon is now thought to have formed from the debris of a giant impact. The bifurcation diagram nonetheless remains the classic example of how a symmetry can be broken as a single parameter, here the angular momentum, is increased.

Haumea. Are there any real bodies that resemble Jacobi ellipsoids? For a planet like the Earth $\Omega^2/\pi G\rho$ is of order 10^{-3} , far below the bifurcation, but in the outer Solar System there is a striking example. The dwarf planet Haumea, which orbits beyond Neptune, rotates once every 3.9 hours, faster than any other large body in the Solar System, and a stellar occultation observed in 2017 showed it to be a triaxial ellipsoid with axes of about 2322, 1704 and 1138 km (Ortiz et al. 2017). As Fig. 4 shows, these axes are almost exactly those of the Jacobi ellipsoid with the same ratio of equatorial axes. Its measured mean density of about 1885 kg m^{-3} gives $\Omega^2/\pi G\rho \approx 0.50$, however, which exceeds the largest rotation rate that any homogeneous equilibrium figure can sustain, and so Haumea cannot be a homogeneous fluid body in equilibrium: it is thought instead to have a denser rocky core beneath an icy mantle, for which the equilibrium figures differ from the classical ones. Its shape is thus a reminder both of the reality of the Jacobi figures and of the importance of internal structure in interpreting the shapes of planets.

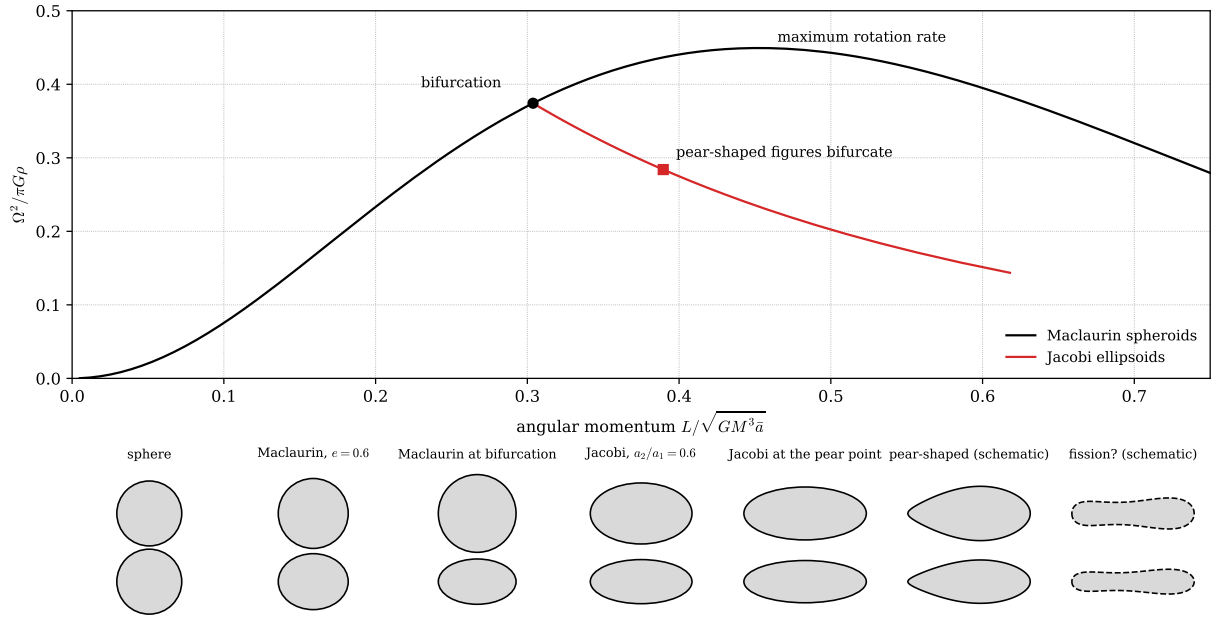


Fig. 3: The Maclaurin and Jacobi sequences of equilibrium figures of a homogeneous rotating fluid, shown as the dimensionless rotation rate $\Omega^2 / \pi G \rho$ against the dimensionless angular momentum, where $\bar{a}$ is the mean radius. The Jacobi sequence branches from the Maclaurin sequence at the point marked, and the pear-shaped figures of Poincaré branch from the Jacobi sequence at the square. Below are the equatorial (upper) and meridional (lower) sections of some members of the sequences, all with the same volume, followed by sketches of a pear-shaped figure and of the fissioning figure that was once imagined to lie beyond it; no closed form exists for these, and the sketches are only schematic.

Maclaurin spheroid at bifurcation
1976 × 1976 × 1152 km

Jacobi ellipsoid, $a_2/a_1 = 0.73$
2319 × 1701 × 1140 km

Haumea
2322 × 1704 × 1138 km

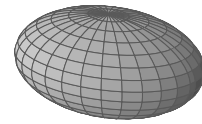
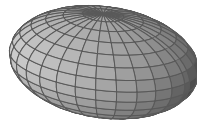
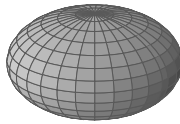


Fig. 4: The Maclaurin spheroid at the point of bifurcation, the Jacobi ellipsoid whose equatorial axes are in the ratio 0.73, and the dwarf planet Haumea as measured by Ortiz et al. (2017), all drawn to the same scale and with the same volume as Haumea.

What you need to know and be able to do

- (i) Know the form of the equilibrium equations, and that they hold in a general planet and not just an elastic one.
- (ii) That the equilibrium equations are ~~under-determined, but they admit non-unique solutions subject to the two compatibility conditions~~ underdetermined, and why.
- (iii) What a hydrostatic equilibrium state is, and be able to derive the constraints imposed on the level surfaces of p , ρ , and γ .
- (iv) Derive the hydrostatic equilibrium state in a non-rotating planet.
- (v) Know in outline how solutions can be obtained within a slowly rotating planet. Here you will not need to reproduce the full derivation, but may be asked to perform related calculations with appropriate guidance.
- ~~(vi) How minimum stress fields can be defined and their relation to slow viscous flow. Again, you do not need to learn this derivation in full, but may be asked to perform related calculations given the necessary information.~~